CS 171: Problem Set 7

Due Date: Saturday, November 14, 2026 at 8:59 PM via Gradescope

1 Aggregate Signatures (15 points)

Let $(\mathbb{G}, \mathbb{G}_T, q, g, e)$ be bilinear-group parameters, where $\mathbb{G}$ and $\mathbb{G}_T$ have prime order q, g generates $\mathbb{G}$, and $e : \mathbb{G} \times \mathbb{G} \to \mathbb{G}_T$ is an efficiently computable, nondegenerate bilinear map. Let $H : \{0,1\}^* \to \mathbb{G}$ be a random oracle. All users share these parameters and H.

Consider the following signature scheme $\Sigma = (\mathsf{Gen}, \mathsf{Sign}, \mathsf{Verify})$:

- $\mathsf{Gen}(1^n)$: Sample $x \leftarrow \mathbb{Z}_q$ and output $pk = h = g^x$ and $sk = x$.
- $\mathsf{Sign}(x, m)$: Output $\sigma = H(m)^x$.
- $\mathsf{Verify}(h, m, \sigma)$: Accept if $\sigma \in \mathbb{G}$ and $e(g, \sigma) = e(h, H(m))$; otherwise reject.

Assume that Σ is existentially unforgeable under chosen-message attacks.

Alice and Bob have public keys $h_A = g^a$ and $h_B = g^b$. They sign the same message m, producing $\sigma_A = H(m)^a$ and $\sigma_B = H(m)^b$. Their *aggregate signature* is the single group element

$$\sigma = \sigma_A \sigma_B.$$

(a) **(4 points)** Give a verification algorithm for an aggregate signature, given h_A, h_B, m, σ, using two evaluations of the bilinear map. Prove correctness.

(b) **(7 points)** Suppose Alice's and Bob's key pairs are generated honestly and independently. An adversary receives h_A, h_B and Bob's secret key b, and may query Alice's signing oracle. It wins if it outputs an accepted aggregate signature on a message that it never queried to Alice's signing oracle. Prove that every PPT adversary wins with negligible probability. Give a reduction to the unforgeability of Σ.

(c) **(4 points)** Now suppose the adversary may choose $h_B \in \mathbb{G}$ after receiving Alice's public key h_A, without providing the corresponding secret key. Prove that with this modification, the aggregate signature scheme is insecure.

2 Key Exposure in a Diffie–Hellman Ratchet (15 points)

Let $\mathbb{G}$ be a cyclic group of prime order q with generator g, in which the DDH assumption holds. Consider five successive public values in a Diffie–Hellman ratchet:

$$A_1 = g^{a_1}, \quad A_2 = g^{a_2}, \quad A_3 = g^{a_3}, \quad A_4 = g^{a_4}, \quad A_5 = g^{a_5},$$

where $a_1, \ldots, a_5 \leftarrow \mathbb{Z}_q$ are independent and uniform. The four successive keys are

$$K_{12} = g^{a_1 a_2}, \qquad K_{23} = g^{a_2 a_3}, \qquad K_{34} = g^{a_3 a_4}, \qquad K_{45} = g^{a_4 a_5}.$$

A passive adversary records all five public values and later learns a_3. No other exponents or stored keys are exposed.

(a) **(5 points)** Which of the four keys can the adversary compute? Give explicit computations for the exposed keys.

(b) **(10 points)** Let

$$V = (A_1, A_2, A_3, A_4, A_5, a_3, K_{23}, K_{34}).$$

Prove that

$$(V, K_{12}, K_{45}) \approx_c (V, U_1, U_2),$$

where U_1, U_2 are independent uniform elements of $\mathbb{G}$, independent of V. Give a reduction to DDH.